Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (`NTL-LTER_Lake_ChemistryPhysics_Raw.csv`). Set date columns to date objects.

2. Build a ggplot theme and set it as your default theme.

```
#1

library(tidyverse);library(here);library(cowplot);library(ggplot2);
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## here() starts at /home/guest/EDE_Fall2025
##
##
```

```
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
library(ggthemes);library(agricolae)
```

```
##
## Attaching package: 'ggthemes'
##
## The following object is masked from 'package:cowplot':
##
##     theme_map
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Lake_Chemistry <- read.csv(
  file = here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
  stringsAsFactors = TRUE
)

class(Lake_Chemistry$sampledate)
```

```
## [1] "factor"
```

```
Lake_Chemistry$sampledate <- mdy(Lake_Chemistry$sampledate)

#2

mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "navy"),
        legend.position = "top")
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: The difference between the sample mean at varying depths is 0. Ha: The difference is not 0, it is greater than or less than 0.

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

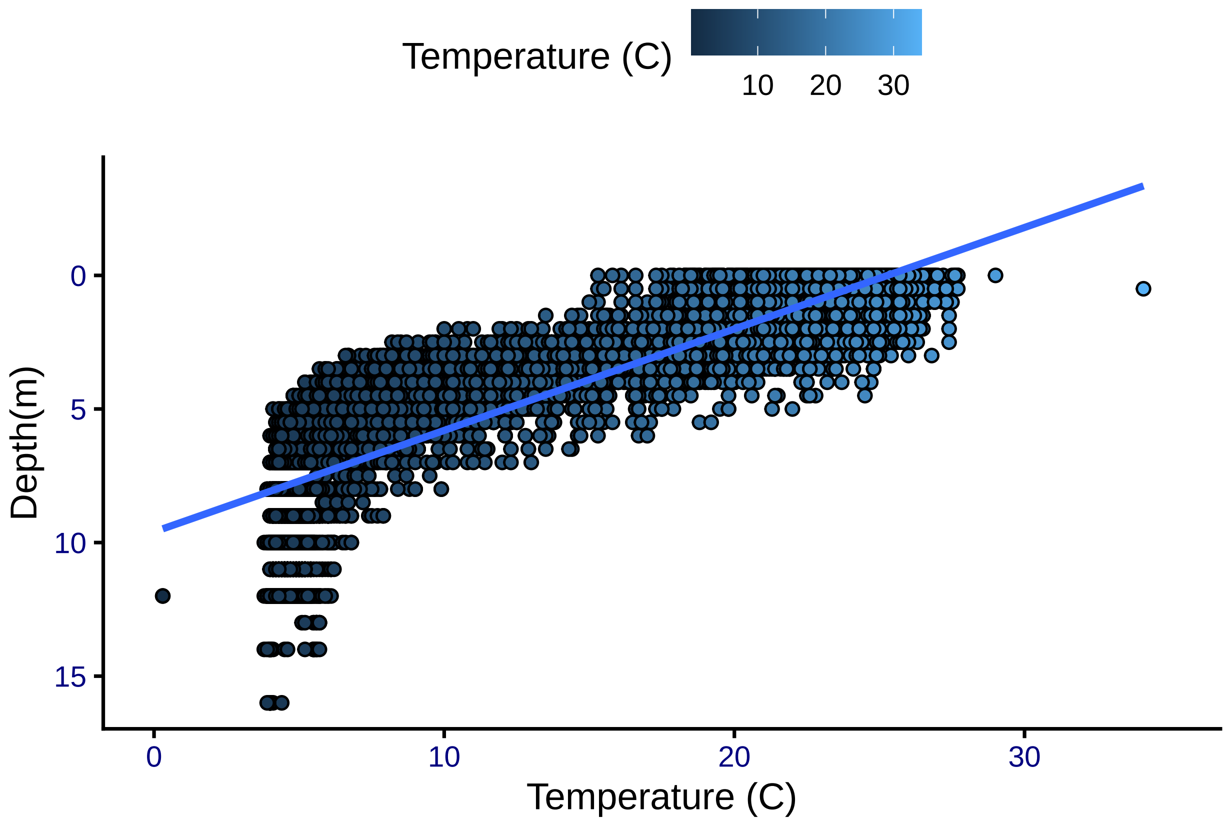
- Only dates in July.
- Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`

- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
 Lake_Chemistry_Sub <- Lake_Chemistry %>%
  filter(daynum >= 183 & daynum <= 213) %>%
  select(c(lakename, year4, daynum, depth, temperature_C)) %>%
  na.omit()

#5

temp_depth_plot <- ggplot(Lake_Chemistry_Sub,
                          aes(x = temperature_C,
                              y = depth,
                              fill = temperature_C)) +
  geom_point(shape = 21) +
  geom_smooth(method = "lm") +
  scale_y_reverse() +
  xlim(0,35) +
  labs(x= "Temperature (C)", y="Depth(m)")+
  scale_fill_continuous(name = "Temperature (C)")

print(temp_depth_plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: The following aesthetics were dropped during statistical transformation: fill.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

   Answer: Based on the slope of the lm line, it shows a strong correlation where as depth increases, temperature decreases. The point distribution echo the thermocline likely present in the lake.

7. Perform a linear regression to test the relationship and display the results.

```r
#7

temp_depth_regression <- lm(data = Lake_Chemistry_Sub, temperature_C ~ depth)

summary(temp_depth_regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = Lake_Chemistry_Sub)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.5606 -3.0380  0.0872  2.9872 13.4706
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 21.98318    0.06840   321.4   <2e-16 ***
## depth       -1.94086    0.01179  -164.7   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.852 on 9671 degrees of freedom
## Multiple R-squared:  0.7371, Adjusted R-squared:  0.7371
## F-statistic: 2.712e+04 on 1 and 9671 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

   Answer:We would expect temperture to drop by -1.94C for each 1m change in depth, as represented by the slope of the regression line. This is based on 9,671 degrees of freedom and the R-square value shows that 73.7% of variation in temperature is explained by the depth of the measurement. The p-value is very small, so we will reject the null hypothesis and say there is a statistically significant relationship between depth and temperature.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.

10. Run a multiple regression on the recommended set of variables.

```r
#9

Temp_AIC <- lm(data = Lake_Chemistry_Sub, temperature_C ~ depth + year4 + daynum)
step(Temp_AIC)
```

```
## Start:  AIC=25998.22
## temperature_C ~ depth + year4 + daynum
##
##          Df Sum of Sq    RSS   AIC
## <none>                142056 25998
## - year4   1       201 142257 26010
## - daynum  1      1237 143293 26080
## - depth   1    402549 544605 38995

##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = Lake_Chemistry_Sub)
##
## Coefficients:
## (Intercept)        depth        year4       daynum
##   -18.19700     -1.94133      0.01611      0.04024
```

```r
#10

Temp_model <- lm(data = Lake_Chemistry_Sub, temperature_C ~ depth + year4 + daynum)
summary(Temp_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = Lake_Chemistry_Sub)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.6857 -3.0267  0.1055  2.9937 13.6038
##
## Coefficients:
##               Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -18.196998   8.741236   -2.082 0.037392 *
## depth        -1.941328   0.011728 -165.528  < 2e-16 ***
## year4         0.016113   0.004353    3.701 0.000216 ***
## daynum        0.040237   0.004385    9.176  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.833 on 9669 degrees of freedom
## Multiple R-squared:  0.7398, Adjusted R-squared:  0.7397
## F-statistic:  9162 on 3 and 9669 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

    Answer: The AIC doesn't recommend removing any variables from the regression. The new model explains 73.9% of variance now (an improvement from 73.7%). This is a small improvement from the singular variable regression.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

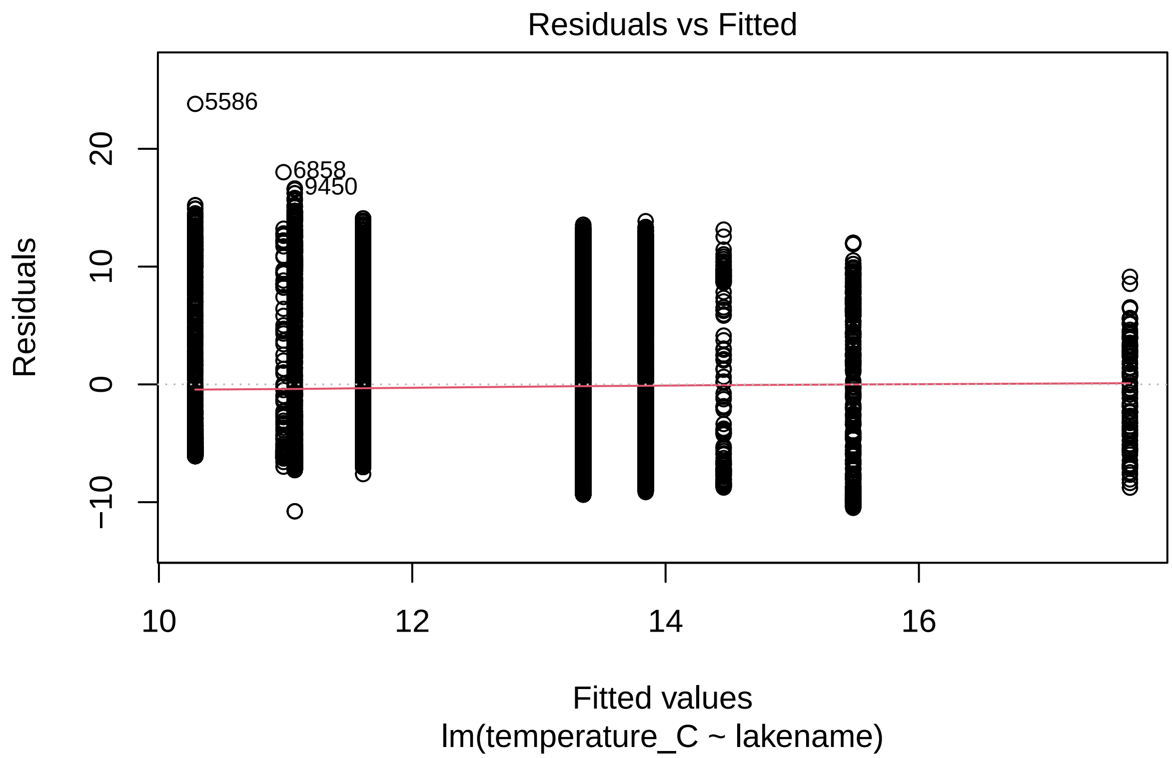
```r
#12
# Format ANOVA as aov
Lake_Chemistry_ANOVA <- aov(data = Lake_Chemistry_Sub, temperature_C ~ lakename)
summary(Lake_Chemistry_ANOVA)
```

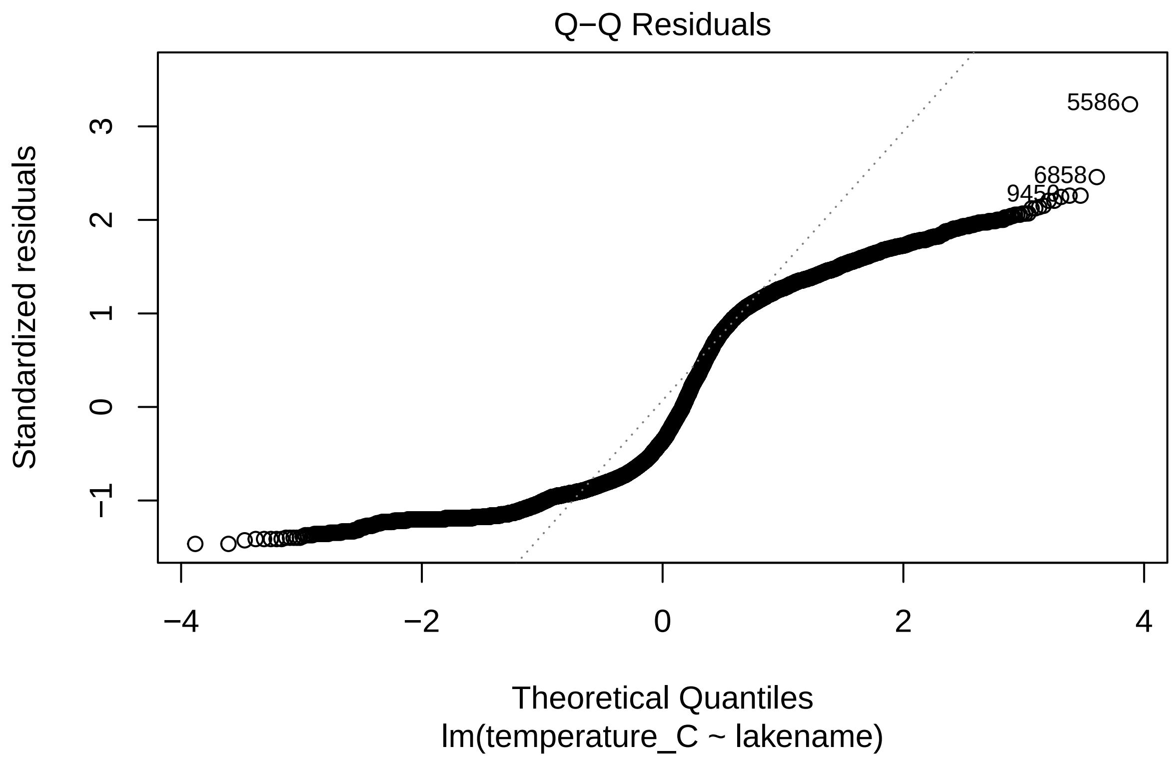
```
##               Df Sum Sq Mean Sq F value Pr(>F)
## lakename       8  22188  2773.5   51.18 <2e-16 ***
## Residuals   9664 523706    54.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

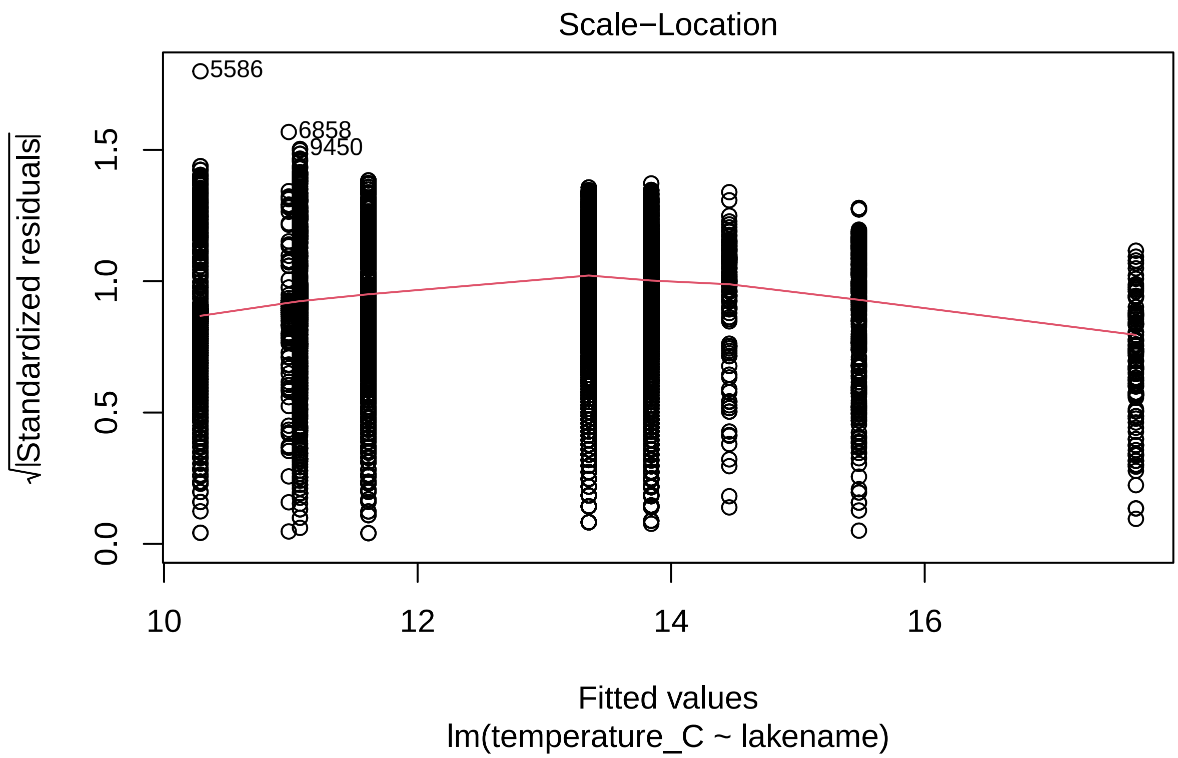
```r
# Format ANOVA as lm
Lake_Chemistry_ANOVA2 <- lm(data = Lake_Chemistry_Sub, temperature_C ~ lakename)
summary(Lake_Chemistry_ANOVA2)
```

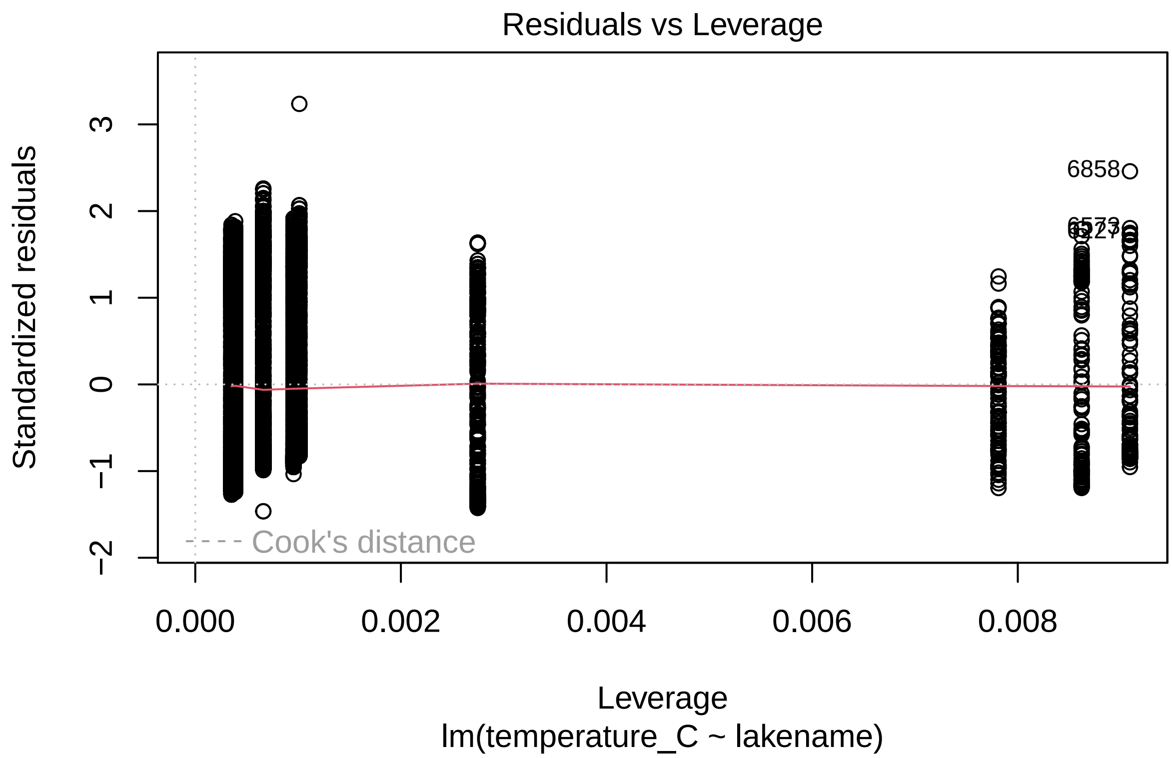
```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = Lake_Chemistry_Sub)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.773  -6.612  -2.673   7.657  23.813
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               17.6664     0.6507  27.151  < 2e-16 ***
## lakenameCrampton Lake     -2.1851     0.7565  -2.889 0.003879 **
## lakenameEast Long Lake    -7.3795     0.6915 -10.671  < 2e-16 ***
## lakenameHummingbird Lake  -6.6828     0.9571  -6.982 3.09e-12 ***
## lakenamePaul Lake         -3.8234     0.6666  -5.735 1.00e-08 ***
## lakenamePeter Lake        -4.3162     0.6652  -6.489 9.08e-11 ***
## lakenameTuesday Lake      -6.5937     0.6777  -9.730  < 2e-16 ***
## lakenameWard Lake         -3.2078     0.9437  -3.399 0.000679 ***
## lakenameWest Long Lake    -6.0542     0.6893  -8.783  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.361 on 9664 degrees of freedom
## Multiple R-squared:  0.04064,    Adjusted R-squared:  0.03985
## F-statistic: 51.18 on 8 and 9664 DF,  p-value: < 2.2e-16
```

```r
plot(Lake_Chemistry_ANOVA2)
```


Residuals vs Fitted
5586
6858
9450
Residuals
20
10
0
−10
10
12
14
16
Fitted values
lm(temperature_C ~ lakename)


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(temperature_C ~ lakename)
5586
6858
9450
−4
−2
0
2
4
−1
0
1
2
3


Scale−Location
5586
6858
9450
√|Standardized residuals|
0.0
0.5
1.0
1.5
10
12
14
16
Fitted values
lm(temperature_C ~ lakename)



13. Is there a significant difference in mean temperature among the lakes? Report your findings.

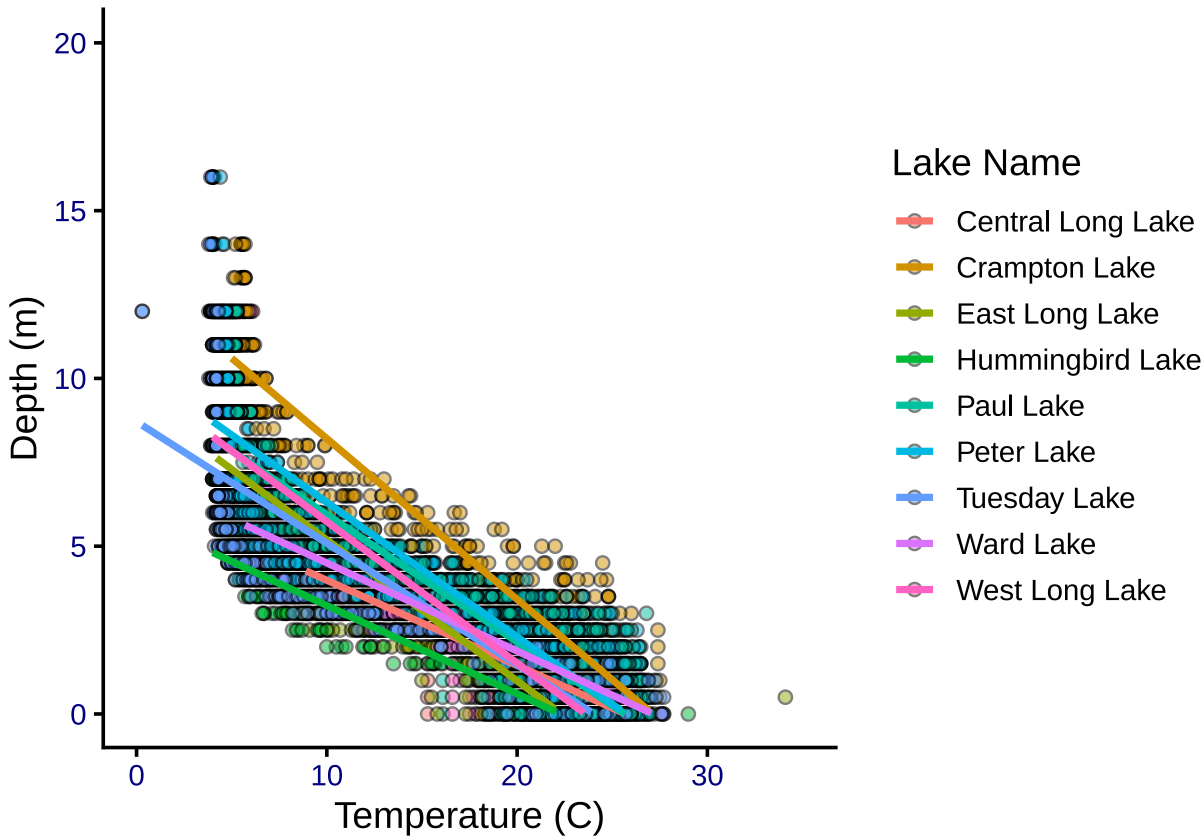
    Answer: Yes, the R-squared value is smaller than 0.05, so we reject the null hypothesis and see that there is a significant difference in mean temperature.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a geom_smooth (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.

temp_depth_plot2 <- ggplot(Lake_Chemistry_Sub, aes(x = temperature_C,
                                                    y = depth,
                                                    fill = lakename)) +
  geom_point(shape = 21, alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, aes(color = lakename)) +
  labs(x = "Temperature (C)", y = "Depth (m)", color = "Lake Name",
       fill = "Lake Name") +
  xlim(0,35) +
  ylim(0,20) +
  theme(legend.position = "right")

print(temp_depth_plot2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 94 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
TukeyHSD(Lake_Chemistry_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = Lake_Chemistry_Sub)
##
## $lakename
##                                           diff        lwr        upr     p adj
## Crampton Lake-Central Long Lake    -2.18508757 -4.5319912  0.1618160 0.0915179
## East Long Lake-Central Long Lake   -7.37946293 -9.5249061 -5.2340198 0.0000000
## Hummingbird Lake-Central Long Lake -6.68276989 -9.6520798 -3.7134600 0.0000000
## Paul Lake-Central Long Lake        -3.82336936 -5.8915944 -1.7551443 0.0000004
## Peter Lake-Central Long Lake       -4.31624752 -6.3799766 -2.2525184 0.0000000
## Tuesday Lake-Central Long Lake     -6.59373914 -8.6961647 -4.4913136 0.0000000
## Ward Lake-Central Long Lake        -3.20778556 -6.1355040 -0.2800671 0.0195251
## West Long Lake-Central Long Lake   -6.05416916 -8.1927792 -3.9155591 0.0000000
## East Long Lake-Crampton Lake       -5.19437536 -6.5946962 -3.7940545 0.0000000
## Hummingbird Lake-Crampton Lake     -4.49768232 -6.9825914 -2.0127732 0.0000007
```

```
## Paul Lake-Crampton Lake           -1.63828179 -2.9171590 -0.3594045 0.0023129
## Peter Lake-Crampton Lake          -2.13115995 -3.4027534 -0.8595665 0.0000072
## Tuesday Lake-Crampton Lake        -4.40865157 -5.7421303 -3.0751729 0.0000000
## Ward Lake-Crampton Lake           -1.02269799 -3.4577560  1.4123600 0.9307880
## West Long Lake-Crampton Lake      -3.86908159 -5.2589107 -2.4792525 0.0000000
## Hummingbird Lake-East Long Lake    0.69669304 -1.5988991  2.9922852 0.9905616
## Paul Lake-East Long Lake           3.55609357  2.7014024  4.4107847 0.0000000
## Peter Lake-East Long Lake          3.06321541  2.2194620  3.9069688 0.0000000
## Tuesday Lake-East Long Lake        0.78572379 -0.1486934  1.7201409 0.1828556
## Ward Lake-East Long Lake           4.17167737  1.9301428  6.4132120 0.0000003
## West Long Lake-East Long Lake      1.32529377  0.3120836  2.3385039 0.0016418
## Paul Lake-Hummingbird Lake         2.85940053  0.6358062  5.0829949 0.0021745
## Peter Lake-Hummingbird Lake        2.36652237  0.1471092  4.5859355 0.0263810
## Tuesday Lake-Hummingbird Lake      0.08903074 -2.1664094  2.3444709 1.0000000
## Ward Lake-Hummingbird Lake         3.47498433  0.4355186  6.5144501 0.0117238
## West Long Lake-Hummingbird Lake    0.62860073 -1.6606065  2.9178079 0.9952002
## Peter Lake-Paul Lake              -0.49287816 -1.1146082  0.1288519 0.2521516
## Tuesday Lake-Paul Lake            -2.77036979 -3.5104805 -2.0302590 0.0000000
## Ward Lake-Paul Lake                0.61558380 -1.5521583  2.7833259 0.9939613
## West Long Lake-Paul Lake          -2.23079980 -3.0681906 -1.3934090 0.0000000
## Tuesday Lake-Peter Lake           -2.27749162 -3.0049438 -1.5500394 0.0000000
## Ward Lake-Peter Lake               1.10846196 -1.0549910  3.2719149 0.8108720
## West Long Lake-Peter Lake         -1.73792164 -2.5641457 -0.9116976 0.0000000
## Ward Lake-Tuesday Lake             3.38595358  1.1855572  5.5863500 0.0000641
## West Long Lake-Tuesday Lake        0.53956999 -0.3790495  1.4581895 0.6673292
## West Long Lake-Ward Lake          -2.84638360 -5.0813788 -0.6113884 0.0025399
```

```
Lake_Chemistry_Groups <- HSD.test(Lake_Chemistry_ANOVA, "lakename", group = TRUE)
Lake_Chemistry_Groups
```

```
## $statistics
##    MSerror   Df     Mean       CV
##   54.19142 9664 12.74849 57.74396
##
## $parameters
##    test   name.t ntr StudentizedRange alpha
##   Tukey lakename   9          4.38751  0.05
##
## $means
##                     temperature_C      std    r        se Min  Max    Q25   Q50
## Central Long Lake       17.66641 4.196292  128 0.6506692 8.9 26.8 14.400 18.40
## Crampton Lake           15.48132 7.347999  364 0.3858465 5.0 27.5  7.500 17.05
## East Long Lake          10.28694 6.765204  988 0.2341999 4.2 34.1  5.000  6.50
## Hummingbird Lake        10.98364 6.779212  110 0.7018898 4.0 29.0  5.225  7.80
## Paul Lake               13.84304 7.314316 2575 0.1450697 4.7 27.7  6.500 12.40
## Peter Lake              13.35016 7.670045 2835 0.1382575 4.0 26.9  5.600 11.50
## Tuesday Lake            11.07267 7.715792 1511 0.1893795 0.3 27.7  4.400  6.80
## Ward Lake               14.45862 7.409079  116 0.6834964 5.7 27.6  7.200 12.55
## West Long Lake          11.61224 6.989346 1046 0.2276142 4.0 25.7  5.400  8.10
##                        Q75
## Central Long Lake 21.000
## Crampton Lake     22.400
## East Long Lake    16.025
## Hummingbird Lake  16.600
```

```
## Paul Lake         21.400
## Peter Lake        21.500
## Tuesday Lake      19.400
## Ward Lake         23.200
## West Long Lake    18.800
##
## $comparison
## NULL
##
## $groups
##                     temperature_C groups
## Central Long Lake        17.66641      a
## Crampton Lake            15.48132     ab
## Ward Lake                14.45862     bc
## Paul Lake                13.84304      c
## Peter Lake               13.35016      c
## West Long Lake           11.61224      d
## Tuesday Lake             11.07267     de
## Hummingbird Lake         10.98364     de
## East Long Lake           10.28694      e
##
## attr(,"class")
## [1] "group"
```

16.From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

> Answer:Based on the groups, Paul and Ward Lakes have the same mean temperature. There is no lake that does not share a group with at least one other lake, so there are no statistically distinct lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

    Answer: We could also use a two-sample T test to compare the temperature means for the two lakes.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```r
Lake_Chemistry_CW <- Lake_Chemistry_Sub %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

twosample <- t.test(Lake_Chemistry_CW$temperature_C ~ Lake_Chemistry_CW$lakename)
twosample
```

```
##
##  Welch Two Sample t-test
##
## data:  Lake_Chemistry_CW$temperature_C by Lake_Chemistry_CW$lakename
```

```
## t = 1.2972, df = 192.4, p-value = 0.1961
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is n
## 95 percent confidence interval:
##  -0.5323014  2.5776973
## sample estimates:
## mean in group Crampton Lake     mean in group Ward Lake
##                    15.48132                    14.45862
```

Answer: The test shows that the means are the same, so we accept the null hypthesis. This makes sense as the two lakes shared a group in part 16, so we would expect them to have statistically similar temperatures. We knoe this because the p-value is >0.05.